Unified FWA Field Framework:
Gravity and Junction Dynamics

Abstract

Fractal–Wave Analysis (FWA) is formulated as a single nonlinear field framework governed by a universal operator E_{FWA}. Two independent projections of this operator describe phenomena at radically different scales: (1) galactic dynamics, where FWA yields the observed logarithmic gravitational potential without invoking dark matter; and (2) baryon–junction dynamics, where FWA represents the experimentally observed Y–topology as a coherent three–branch wave state with a calculable transport eigenvalue. The unified structure demonstrates that both macroscopic and microscopic behaviors can arise from the same wave–operator formalism. The framework is falsifiable: its predictions must be computed from the operator itself, not fitted to data.

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1. Fundamental Equation

FWA is defined by the nonlinear field equation:

$$E_{FWA}[\Psi] = 0$$

where Ψ denotes the relevant wave configuration. Different physical regimes correspond to different projections of the same operator. This paper focuses on two projections:

- **FWA Gravity** — large–scale dynamics of galaxies,
- **FWA Junction** — coherent three–branch baryon structures.

Both projections follow from the same mathematical foundation.

2. FWA Gravity Projection

2.1 Newtonian Limit and Operator Modification

In the Newtonian limit of General Relativity,

$$\nabla^2 \Phi = 4\pi G \rho_b$$

which yields the Keplerian velocity profile

$$v^2(r) = GM / r$$

Thus GR predicts $v(r) \propto r^{-1/2}$ at large radii. However, spiral galaxies exhibit flat rotation curves:

$$v(r) \to v_0 \quad (r \to \infty)$$

implying

$$d\Phi / dr = v_0^{\,2} / r \;\Rightarrow\; \Phi(r) \sim v_0^{\,2} \ln r$$

This logarithmic potential is incompatible with GR + baryons alone.

FWA resolves the discrepancy by modifying the operator:

$$\nabla^2 \Phi - 4\pi G \rho_b - \Delta_{FWA}[\Phi] = 0$$

where Δ_{FWA} is the asymptotic wave–operator term. Its large–scale behavior yields $\Phi(r) \sim v_0^{\,2} \ln r$ without introducing dark matter.

2.2 Effective Density Interpretation

Substituting the logarithmic potential into the Poisson operator gives

$$\nabla^2 \Phi \sim v_0^{\,2} / r^2$$

corresponding to an effective density

$$\rho_{eff}(r) = v_0^{\,2} / (4\pi G r^2)$$

In GR this term is interpreted as a dark–matter halo. In FWA it arises naturally from the operator Δ_{FWA}, not from additional matter.

Thus the gravitational sector illustrates the core FWA principle: modify the operator, not the mass content.

3. FWA Junction Projection

3.1 Coherent Three–Branch Wave State

A baryon junction is represented as a coherent three–branch wave configuration:

$$\Psi_J = \Sigma_{k=1..3} A_k e^{i\phi_k}, \quad \phi_k = 2\pi k/3 + \delta\phi_k$$

The unperturbed phases satisfy the root–of–unity cancellation:

$$\Sigma_{k=1..3} e^{i2\pi k/3} = 0$$

interpreted in FWA as a dynamic–zero condition of a perfectly balanced junction. This representation does not replace SU(3) color; it provides an independent wave–based description of the observed Y–topology.

3.2 Symmetry–Error Functional

Departures from perfect symmetry are quantified by

$$Q_J^{\,2} = |\, \Sigma_{k=1..3} A_k e^{i\phi_k} \,|^2$$

For equal amplitudes $A_k = 1$ and small distortions $\delta\phi_k$, $e^{i\delta\phi_k} \approx 1 + i\delta\phi_k$, and the zeroth–order term cancels. The leading contribution is quadratic:

$$Q_J^{\,2} \propto |\, \Sigma_{k=1..3} \delta\phi_k e^{i2\pi k/3} \,|^2$$

This is a concrete mathematical prediction: junction symmetry errors scale as $\delta\phi^2$.

3.3 Dynamical Recursion and Transport Law

The junction evolves under the general FWA recursion

$$\Psi_{n+1} = F(\Psi_n, V_n, d_n, T_n)$$

where T encodes the local medium state. Linearization around a stationary junction yields $\Psi_{n+1} \approx \lambda_J \Psi_n$, with λ_J the dominant eigenvalue.

Over a rapidity interval Δy, $C(\Delta y) = e^{-\lambda_J \Delta y}$. This leads to the transport law:

$$dN_B / dy = N_0 \exp[-\lambda_J(y - y_0)]$$

The eigenvalue λ_J must be computed from the FWA equations, not fitted to STAR data. Agreement supports the model; disagreement falsifies it.

4. Unified Structure

Both sectors are projections of the same fundamental operator $E_{FWA}[\Psi] = 0$:

Galactic scales: $E_{grav}[\Phi, \rho_b] = 0$
Hadronic scales: $E_{junction}[\Psi_J, T] = 0$

- **Gravity:** FWA modifies the operator to produce the observed logarithmic potential without dark matter.
- **Junctions:** FWA represents Y–topology as a coherent wave state with a calculable transport eigenvalue.

Both follow the same scientific principle: A physical explanation becomes a theory only when its mathematical structure produces consequences that nature can contradict.

5. Final Consolidated Block

(1) Fundamental field equation:

$E_{FWA}[\Psi] = 0$

(2) Gravity projection:

$\nabla^2 \Phi - 4\pi G\rho_b - \Delta_{FWA}[\Phi] = 0$

$\Phi(r) \sim v_0^2 \ln r, \quad v(r) \sim v_0$

(3) Junction projection:

$\Psi_J = \Sigma A_k e^{i\phi_k}, \quad Q_J^2 = |\Sigma A_k e^{i\phi_k}|^2$

$\Psi_{n+1} \approx \lambda_J \Psi_n$

$dN_B/dy = N_0 e^{-\lambda_J(y-y_0)}$